ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 2

ELECTRONICS AND DIGITAL CONCEPTS FOR ROBOTICS

PRACTICE EXAM 2

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 2. Section A: objective questions (30) · Section B: SI units and atomic theory, electrical supply systems and Joule's law, resistor colour codes and series-parallel circuits, cells and batteries, diodes and semiconductors, circuit symbols, soldering and testing components (70).

Written by Academic Central for practice, following the NC(V) subject guidelines for NC(V) Level 2. It is not a Department of Higher Education and Training examination paper.

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Show ALL formulae, substitutions and calculations, and write units with all answers. Round off to TWO decimal places.
5. A non-programmable calculator may be used.
6. Write neatly and legibly.

SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10).

1.1.1 The SI unit of electrical charge is the …

A ampere. B volt. C joule. D coulomb.

1.1.2 The prefix "milli" means …

A $\times 10^{3}$ B $\times 10^{-3}$ C $\times 10^{-6}$ D $\times 10^{6}$

1.1.3 The particles that move to form an electric current in a metal wire are …

A protons. B neutrons.
C free electrons. D nuclei.

1.1.4 Silicon is an example of a/an …

A semiconductor. B conductor. C insulator. D magnet.

1.1.5 The heat produced in a resistor is calculated with …

A Newton's law. B Joule's law.
C Pascal's law. D Hooke's law.

1.1.6 A resistor with the colour bands brown, black, red has a value of …

A 102 Ω B 10 Ω C 12 kΩ D 1 kΩ

1.1.7 Adding a small amount of phosphorus to silicon produces …

A N-type material. B P-type material.
C an insulator. D a superconductor.

1.1.8 A diode allows current to flow …

A in both directions equally. B only when it is cold.
C in one direction only. D only in AC circuits.

1.1.9 Electricity in South Africa is transmitted over long distances at very high voltage mainly to …

A make it safer at home.
B change it to DC.
C reduce power losses in the lines.
D increase the current.

1.1.10 The best tool for removing excess solder from a joint is a …

A solder sucker (desoldering pump).
B wire stripper.
C side cutter.
D multimeter.

(10)

1.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'true' or 'false' next to the question number (1.2.1–1.2.10).

1.2.1 The nucleus of an atom contains protons and neutrons.

1.2.2 4,7 kΩ is equal to 470 Ω.

1.2.3 Total resistance in series is the sum of the individual resistances.

1.2.4 Cells connected in parallel give a higher voltage than one cell.

1.2.5 The cathode of a diode is marked with a band.

1.2.6 A forward-biased silicon diode has a voltage drop of about 0,7 V.

1.2.7 Veroboard (stripboard) is used to build permanent soldered circuits.

1.2.8 A soldering iron should be put down on the workbench when it is not being used.

1.2.9 A good diode shows a low reading one way and a high (open) reading the other way on a multimeter diode test.

1.2.10 Eskom's power stations generate direct current for homes.

(10)

1.3 Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (1.3.1–1.3.10).

	COLUMN A		**COLUMN B**
1.3.1	Positively charged particle in the nucleus	A	PN junction
1.3.2	Electrons in the outer shell of an atom	B	transformer
1.3.3	Semiconductor with extra holes	C	tolerance
1.3.4	Boundary between P-type and N-type material	D	P-type
1.3.5	Fourth band on a resistor	E	national grid
1.3.6	Unit of electrical energy used on electricity bills	F	solder
1.3.7	Network of power lines that carries electricity across the country	G	valence electrons
1.3.8	Changes the voltage between transmission and distribution	H	flux
1.3.9	Metal alloy melted to join components	I	kilowatt-hour
1.3.10	Substance that cleans the joint so solder flows well	J	proton

(10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2: SI UNITS, ATOMS AND SUPPLY SYSTEMS

2.1 Convert the following: (a) 2 200 Ω to kΩ; (b) 0,045 A to mA; (c) 3,3 MΩ to Ω; (d) 15 000 µF to F. (4)

2.2 Draw a labelled diagram of a silicon atom (14 electrons) showing the nucleus and the electron shells. (4)

2.3 Explain, in terms of free electrons, why copper is a good conductor and rubber is a good insulator. (4)

2.4 Describe the path of electricity from a coal power station to a house, naming the main stages. (4)

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QUESTION 3: OHM'S LAW, JOULE'S LAW AND NETWORKS

3.1 A soldering iron is rated 230 V, 40 W.

3.1.1 Calculate the current it draws. (2)

3.1.2 Calculate the resistance of its heating element. (2)

3.1.3 Calculate the heat energy (in joules) it produces in 10 minutes. (3)

3.1.4 Calculate the cost of using it for 30 hours if electricity costs R3,20 per kWh. (3)

3.2 A 6 Ω resistor is connected in series with a parallel combination of a 12 Ω and a 4 Ω resistor. The circuit is connected to a 9 V battery.

3.2.1 Calculate the total resistance. (4)

3.2.2 Calculate the total current. (2)

3.2.3 Calculate the voltage across the parallel combination. (2)

[18]

QUESTION 4: COMPONENTS, CELLS AND SEMICONDUCTORS

4.1 Give the colour bands (four bands) of: (a) a 470 Ω, 5% resistor; (b) a 22 kΩ, 10% resistor. (4)

4.2 Six 1,5 V cells, each able to supply 0,5 A, are connected (a) all in series and (b) in two parallel groups of three cells in series. Calculate the voltage and the maximum current for each connection. (6)

4.3 Explain how N-type and P-type semiconductor material is made. (4)

4.4 Draw the symbol of a diode and explain what forward bias and reverse bias mean. (4)

[18]

QUESTION 5: DRAWINGS, SOLDERING AND TESTING

5.1 Draw the circuit symbols for: (a) a fuse; (b) a light-emitting diode; (c) an electrolytic capacitor; (d) a single-pole switch. (4)

5.2 List SIX steps (in order) to make a good soldered joint on veroboard. (6)

5.3 Name TWO safety precautions when soldering. (2)

5.4 Explain how to test whether a resistor is still within its tolerance, using a multimeter. (6)

[18]

TOTAL SECTION B: [70]

TOTAL: 100

END OF QUESTION PAPER